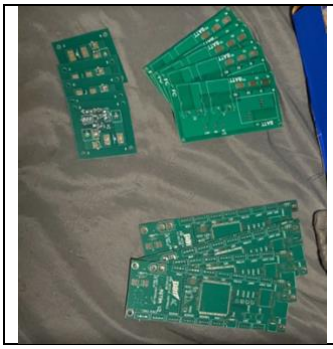
PSPAC

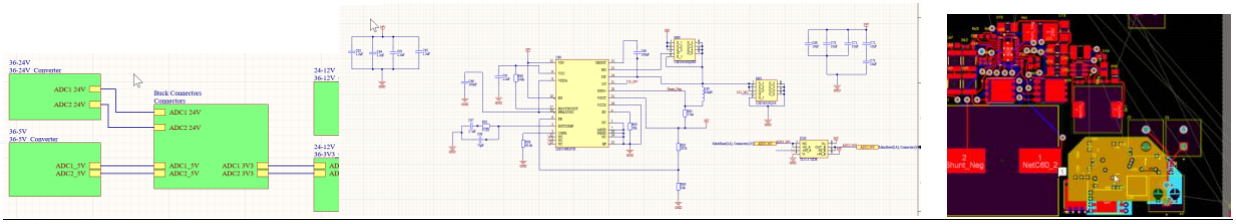
Purdue Space Program Active Controls Avionics Team

	I worked on the Flight Control Board with my former avionics lead Xander. It is built around an STM32H7 dual-core microcontroller for processing. I was focused on implementing the 130 MB of program storage. It interfaces with essential peripherals using SPI, UART, and ADC, enabling fast and reliable communication with onboard systems. He mainly worked on implementing the sensor suite. The top left board which monitors power with a shunt was my starting project before that. I worked on a similar system in Embedded systems club as well.

Current work

Buck PCB Responsible Engineer & Working on Engine Controller

- Buck Layout in progress in Altium (only 24v buck shown for schematic)



SOCET

I am currently also on Purdue's SoCET Team, which does digital design projects. In a team of 3, I've been working on a matrix calculator using SystemVerilog on an FPGA. It can perform matrix operations such as addition, subtraction, and multiplication by leveraging parallelism inherent in digital logic, significantly improving performance compared to software-based approaches. It is modularized, including arithmetic units, control logic, and memory interfaces. Focus was placed on timing optimization and pipelining to ensure reliable operation at higher clock frequencies.

Through SoCET's previous projects, I have also created memory copiers and accelerators (+ driver for the accelerator) in SystemVerilog. I have also started learning Assembly and ASIC design through the program.

First author Vertical Climbing Research Paper

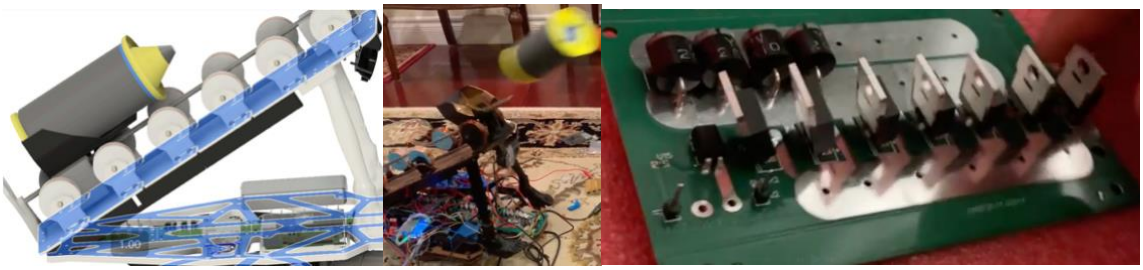
- All mechanical, electrical, and software for robot done by me
- Published in IEEE International Conference on Robotic Computing 2024 and Presented in Hitachi Research Lab in Tokyo, Japan



Electromagnetic Coilgun

https://hyweric.github.io/port/k-capstone/kcapstone (Fun High School Project 12th Sem 2)

- Designed high current supercapacitor discharging PCBs
- Mechanical components also designed by me, though they were kind of rushed as I wanted to focus on the physics and electrical component
- Modified a buck boost converter to be controlled by digital potentiometer through a Raspi to allow for adjustable launch distances

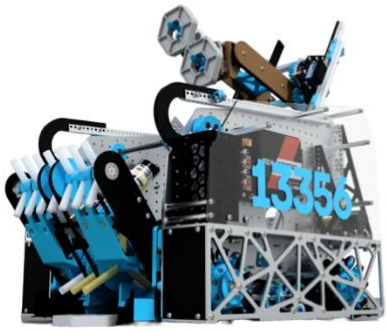


(one of my first PCBs I made)

Hardware Co Captain - FTC Robots

- Led Design and testing mechanical of mechanical subassemblies
- Consistently ranked among the top 30 teams out of 7,000 teams worldwide
- Winning Alliance Captain at the Norcal States
- 3x Maryland Tech Invitational Acceptance (Top 40 teams globally based on robot)

2023-2024 Season Robot:



2022-2023 Season Robot:



2021-2022 Season Robot:



Autodesk Fusion 360 Wire Generator

Creating wires has always been a tedious process in Fusion 360, tediously tracing spline points on 3D sketches. The Fusion 360 Wire Generator Add-on simplifies wire creation in Fusion 360. Simply select start and end points, set parameters, and generate multiple spline options effortlessly. Choose the best fit for the wire and adjust spline points easily.

- 5.0K+ downloads as of 2026



